

where M_{j+1} is the deflated Hamiltonian, based on $M_{j+1} = M_j - \lambda_j |\psi_j\rangle\langle\psi_j|$, where M_j is the deflated Hamiltonian in the previous iteration, with $M_1 := M$ and $|\psi_j\rangle$ is the approximate estimate in (7) based on M_j with corresponding energy $\lambda_j := \langle\psi_j|M_j|\psi_j\rangle$. A generic procedure for solving the multicomponent VQE with deflation is detailed in Algorithm 1.

In practice though, successively reconstructing a deflated Hamiltonian represents a computational burden as it would require the construction of the deflated Hamiltonian—that entails a sum of Pauli operators for measuring it. Existing literature relies on implicit deflation algorithms, where one still uses the VQE framework but changes the objective to favor orthogonal solutions by adding a penalty term for vectors that are not orthogonal to the current vector. Such an example is the case of the Variational Quantum Deflation (VQD) algorithm [30]; we describe and compare with VQD in the experimental section.

In this work, we are exploring a novel penalty term in the objective, inspired by a game-theoretic formulation.

3. An EigenGame for VQE

What is EigenGame? A recent study for calculating PCA in a distributed fashion led to the creation of EigenGame [29]. EigenGame is an algorithm posed as a competitive game, where “players” are assigned to estimate eigenvectors, distinct from other eigenvectors of M . The way the algorithm operates leads to the Nash equilibrium of an eigenvalue game and does so by a sequential or a parallel version of EigenGame, namely EigenGame and EigenGameR [29], respectively.

Let $v_i \in \mathbb{R}^p$ denote an *approximation* of the i -th principal component of M . EigenGame defines an alternative *utility objective* to be maximized by the i -th “player” in this game, defined as follows:

$$\max_{v_i: \|v_i\|_2=1} \left\{ f(v_i \mid v_{j<i}) := v_i^\top M v_i - \sum_{j<i} \frac{(v_i^\top M v_j)^2}{v_j^\top M v_j} \right\}. \quad (4)$$

Algorithm 2 EigenGame for the i -th player

Input: $M \in \mathbb{R}^{p \times p}$, initial vectors $v_{i,\text{init}}$, learned parents $v_{j<i}$, step size α , and # of iters t_i .

$v_{i,1} \leftarrow v_{i,\text{init}}$

for $t = 1 : t_i$ **do**

$$\nabla_{v_i} f(v_{i,t} \mid v_{j<i}) = 2M \left(v_{i,t} - \sum_{j<i} \frac{v_{i,t}^\top M v_j}{v_j^\top M v_j} v_j \right)$$

$$\tilde{v}_{i,t+1} \leftarrow v_{i,t} + \alpha \nabla_{v_i} f(v_{i,t} \mid v_{j<i})$$

$$v_{i,t+1} \leftarrow \frac{\tilde{v}_{i,t+1}}{\|\tilde{v}_{i,t+1}\|_2}$$

end for

Return v_{i,t_i}

The term in **blue** maximizes the variance of the projected data along the vector being optimized. This is analogous to the traditional goal of PCA in (1). The term in **red** penalizes the alignment of the vector with any other vectors that have already been optimized. This term ensures *implicit* orthogonality among the vectors, mimicking the orthogonality constraint in PCA. The intuition behind this redefined objective function is to transform the PCA problem from a passive eigenvalue decomposition into an active, competitive process: each “player”

seeks to maximize its utility in terms of variance captured, considering the presence and position of other vectors. The gradient of the utility function, *assuming access to a first-order oracle*, can be calculated as $\nabla_{v_i} f(v_i \mid v_{j<i}) = 2M \left(v_i - \sum_{j<i} \frac{v_i^\top M v_j}{v_j^\top M v_j} v_j \right)$. We describe EigenGame in Algorithm 2.

Preparing the quantum EigenGame. Our interest steers towards obtaining a utility gradient that uses quantum oracle calls to calculate eigenvalue/eigenvector pairs. We first convert the classical EigenGame objective formulation into a parameterized one, using quantum computing formulations. Denoting the r -th player’s parameters as $\theta^{(r)} \in \mathbb{R}^m$, its corresponding objective becomes:

$$\max_{\theta^{(r)} \in \mathbb{R}^m} \left\{ \langle\psi(\theta^{(r)})|M|\psi(\theta^{(r)})\rangle - \sum_{j<r} \frac{\langle\psi(\theta^{(r)})|M|\psi(\theta^{(j)})\rangle^2}{\langle\psi(\theta^{(j)})|M|\psi(\theta^{(j)})\rangle} \right\} \quad \text{s.t.} \quad |\psi(\theta^{(r)})\rangle = \prod_{i=0}^{\ell-1} U_i(\theta_i^{(r)})|\psi_0\rangle. \quad (5)$$

What differentiates (5) from the classical VQE formulation in (2) is the inclusion of the regularization term in **red**. That is, (5) defines different objectives per “player”. Note that the term in **blue** is